

Lebanese University – Faculty of Engineering III

Sensors & Instrumentation Project

Individual Contribution Report

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Project: Smart Building Security and Fire-Protection System

Contribution Summary

Throughout the development of the Smart Building Security and Fire-Protection System, I actively participated in both the hardware implementation and the project development process.

One of my primary responsibilities was assembling the electronic system by connecting the sensors, modules, and other components to the Arduino using the appropriate wiring. Before the assembly process, I carefully studied and understood the circuit layout designed in Fritzing to ensure that all connections were correctly implemented.

I continuously tested each hardware component after it was added to the circuit to verify its functionality and ensure proper integration with the rest of the system. This testing process helped identify wiring or component issues early and contributed to the overall reliability of the project.

In addition to the hardware work, I collaborated with my teammates in preparing sections of the PowerPoint presentation and the technical report. I also participated in discussing the project design and proposed several important ideas and improvements that were implemented to enhance the system's functionality and overall performance.

During the implementation phase, I assisted in troubleshooting hardware connections, verifying sensor operation, and confirming that the integrated system performed as expected. I also contributed during the final preparation by helping review the presentation materials and ensuring that the project was ready for demonstration.

Throughout the project, I actively participated in team meetings, implementation sessions, and system testing. I contributed to collecting experimental results and organizing the project data. I also helped prepare the project demonstration by supporting the presentation process and ensuring the system was ready for a successful presentation.

Overall, I actively contributed to the hardware assembly, circuit verification, component testing, documentation, presentation preparation, teamwork, and project improvement discussions, helping the team successfully complete the Smart Building Security and Fire-Protection System.

Declaration

I certify that the activities described above accurately reflect my individual contribution to the Smart Building Security and Fire-Protection System project.